Assignment No :2

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Subject Title: COMPUTER PROGRAMMING

Program 1:

Write a program to: • Take 10 integer inputs in an array. • Find and display: Maximum value Minimum value Average

```
#include <iostream>   // Library for input/output
using namespace std;  // Use standard namespace

int main() {
   int arr[10];      // Array to store 10 integers
   int sum = 0;      // Variable to store sum

   cout << "Enter 10 integers:\n";

   // Loop to take input
   for(int i = 0; i < 10; i++) {
      cin >> arr[i];      // Store value in array
      sum += arr[i];      // Add value to sum
   }

   int max = arr[0];  // Assume first value is max
   int min = arr[0];  // Assume first value is min

   // Loop to find max and min
   for(int i = 1; i < 10; i++) {
      if(arr[i] > max)   // If current value is greater
         max = arr[i];  // Update max

      if(arr[i] < min)   // If current value is smaller
         min = arr[i];  // Update min
```

```
    }

    float avg = sum / 10.0;  // Calculate average

    cout << "Maximum = " << max << endl;
    cout << "Minimum = " << min << endl;
    cout << "Average = " << avg << endl;

    return 0;  // End program
}
```

Program 2:

Write a program that: • Uses a function to reverse a 1D array. • Display the array before and after reversal.

```
#include <iostream>
using namespace std;

// Function to reverse array
void reverseArray(int arr[], int size) {
    // Loop runs till half of array
    for(int i = 0; i < size / 2; i++) {
        // Swap first with last
        swap(arr[i], arr[size - i - 1]);
    }
}

int main() {
    int arr[5] = {1, 2, 3, 4, 5};  // Initialize array

    cout << "Before Reverse:\n";

    // Display original array
    for(int i = 0; i < 5; i++)
        cout << arr[i] << " ";

    reverseArray(arr, 5);  // Call function

    cout << "\nAfter Reverse:\n";

    // Display reversed array
    for(int i = 0; i < 5; i++)
        cout << arr[i] << " ";

    return 0;
}
```

↳ Program 3:

Write a program to: • Input two 3 × 3matrices. • Perform matrix addition. • Display the result matrix.

```
   #include <iostream>        // Include library
using namespace std;       // Use standard namespace

int main() {                        // Main function
   int A[3][3], B[3][3], C[3][3];    // Declare three 3x3 matrices

   cout << "Enter first matrix:\n";  // Ask user for first matrix

   for(int i = 0; i < 3; i++)        // Outer loop for rows
      for(int j = 0; j < 3; j++)    // Inner loop for columns
         cin >> A[i][j];           // Input value for matrix A

   cout << "Enter second matrix:\n"; // Ask for second matrix

   for(int i = 0; i < 3; i++)        // Loop rows
      for(int j = 0; j < 3; j++)    // Loop columns
         cin >> B[i][j];           // Input matrix B

   for(int i = 0; i < 3; i++)        // Loop rows
      for(int j = 0; j < 3; j++)    // Loop columns
         C[i][j] = A[i][j] + B[i][j];  // Add matrices

   cout << "Result Matrix:\n";       // Print result message

   for(int i = 0; i < 3; i++) {     // Loop rows
      for(int j = 0; j < 3; j++)   // Loop columns
         cout << C[i][j] << " ";  // Print each element
      cout << endl;               // Move to next line
   }

   return 0;   // End program
}
```

Program 4: Write a program that: • Stores names of 5 students in a string array. • Sorts them in alphabetical order. • Displays the sorted list.

.

```
#include <iostream>          // Include input-output library
#include <string>            // Include string library
using namespace std;         // Use standard namespace

int main() {                 // Main function starts
   string names[5];          // Declare array of 5 strings

   cout << "Enter 5 names:\n";  // Ask user for input

   for(int i = 0; i < 5; i++) { // Loop to input names
      cin >> names[i];          // Store each name
   }

   for(int i = 0; i < 5; i++) {        // Outer loop for sorting
      for(int j = i + 1; j < 5; j++) {// Inner loop for comparison
         if(names[i] > names[j])      // Compare strings alphabetically
            swap(names[i], names[j]); // Swap if out of order
      }
   }

   cout << "Sorted Names:\n";   // Print heading

   for(int i = 0; i < 5; i++) { // Loop to display names
      cout << names[i] << endl; // Print each name
   }

   return 0;   // End program
}
```

- **Program 5:**

  Power System Load Analysis (1D Array + Function) In a power system, hourly load demand (in MW) is recorded. Write a program to: • Store load data for 24 hours in an array. • Create functions to calculate: Total load Peak

```
#include <iostream>        // Include library
using namespace std;       // Use standard namespace

int totalLoad(int arr[], int size) { // Function to calculate total
    int sum = 0;                  // Initialize sum
    for(int i = 0; i < size; i++) { // Loop through array
        sum = sum + arr[i];         // Add each value
    }
    return sum;                   // Return total
}

int peakLoad(int arr[], int size) { // Function to find max value
    int max = arr[0];              // Assume first value is max
    for(int i = 1; i < size; i++) {// Loop from second element
        if(arr[i] > max)           // Check if greater
            max = arr[i];          // Update max
    }
    return max;                   // Return peak value
}

int main() {                      // Main function
    int load[24];                 // Array for 24 hours

    cout << "Enter 24 load values:\n"; // Ask for input

    for(int i = 0; i < 24; i++) { // Loop for input
        cin >> load[i];           // Store each value
    }

    cout << "Total Load = " << totalLoad(load, 24) << endl; //
Call function
    cout << "Peak Load = " << peakLoad(load, 24) << endl;   //
Call function

    return 0;   // End program
}
```

- **Program 6:**

Design a program to: Store student names (string array) and their marks (1D array). Use functions to: Find topper Display pass/fail status Search a student by name

```
#include <iostream>        // Include library
#include <string>          // Include string
using namespace std;       // Use namespace
```

```
int findTopper(int marks[], int size) { // Function to find
topper
   int index = 0;                  // Assume first student is topper
   for(int i = 1; i < size; i++) {    // Loop through marks
      if(marks[i] > marks[index])    // Compare marks
         index = i;                // Update index
   }
   return index;                   // Return topper index
}

void searchStudent(string names[], int marks[], int size,
string key) { // Search function
   for(int i = 0; i < size; i++) {   // Loop through names
      if(names[i] == key) {         // If name matches
         cout << "Found: " << names[i] << " Marks: " <<
marks[i] << endl;
         return;                   // Exit function
      }
   }
   cout << "Student not found\n";   // If not found
}

int main() {                       // Main function
   string names[5];                // Array for names
   int marks[5];                   // Array for marks

   cout << "Enter names and marks:\n"; // Input message

   for(int i = 0; i < 5; i++) {     // Loop for input
      cin >> names[i] >> marks[i]; // Store name and marks
   }

   int topper = findTopper(marks, 5); // Call topper function

   cout << "Topper: " << names[topper] << endl; // Display
topper

   for(int i = 0; i < 5; i++) {     // Loop for pass/fail
      if(marks[i] >= 50)           // Check condition
         cout << names[i] << " Pass\n";
      else
         cout << names[i] << " Fail\n";
   }

   string key;                     // Variable for search
   cout << "Enter name to search: ";
   cin >> key;                     // Input search name

   searchStudent(names, marks, 5, key); // Call search
function

   return 0;   // End program
```

```
}
```

Program 7:

Renewable Energy Monitoring System (1D Array + Functions) A solar power plant records the energy generated (in kWh) every hour for one day.

Write a C++ program to:

• Store energy generation data for 24 hours in a 1D array.

Create functions to:

• Calculate total energy generated in a day

• Find the hour with maximum generation

• Calculate average energy generation

• Display all results clearly.

```
#include <iostream>       // Include library
using namespace std;      // Use namespace

float totalEnergy(float arr[], int size) { // Total energy function
   float sum = 0;                  // Initialize sum
   for(int i = 0; i < size; i++) {      // Loop
      sum += arr[i];               // Add values
   }
   return sum;                     // Return sum
}

int maxHour(float arr[], int size) {      // Function to find max hour
   int index = 0;                  // Assume first index
   for(int i = 1; i < size; i++) {      // Loop
      if(arr[i] > arr[index])         // Compare
         index = i;                // Update index
   }
   return index;                   // Return index
}

float average(float arr[], int size) {      // Average function
   return totalEnergy(arr, size) / size; // Return average
}

int main() {                       // Main function
```

```
  float energy[24];                // Array of 24 hours

  cout << "Enter 24 values:\n";      // Input message

  for(int i = 0; i < 24; i++) {       // Loop
    cin >> energy[i];              // Store values
  }

  cout << "Total = " << totalEnergy(energy, 24) <<
endl;
  cout << "Peak Hour = " << maxHour(energy, 24) <<
endl;
  cout << "Average = " << average(energy, 24) <<
endl;

  return 0;  // End
}
```

↳ program 8:

Battery Health Monitoring (1D Array + Functions) In battery management systems, voltage readings are monitored to ensure safe operation.

Write a program to:

• Store 20 voltage readings of a battery in an array. •

Define safe voltage limits (e.g., 3.0V to 4.2V).

• Create functions to:

• Count how many readings are below or above safe limits •

Find the maximum and minimum voltage •

Display a warning message if unsafe readings exist

```
#include <iostream>       // Include library
using namespace std;      // Use namespace

int main() {              // Main function
  float v[20];            // Array for 20 readings
  float low = 3.0, high = 4.2; // Safe limits

  int below = 0, above = 0; // Counters

  cout << "Enter 20 readings:\n";

  for(int i = 0; i < 20; i++) { // Input loop
    cin >> v[i];            // Store value
```

```
    if(v[i] < low)          // Check below limit
       below++;             // Increment
    else if(v[i] > high)    // Check above limit
       above++;             // Increment
  }

  float max = v[0], min = v[0]; // Initialize max and min

  for(int i = 1; i < 20; i++) { // Loop
    if(v[i] > max) max = v[i]; // Update max
    if(v[i] < min) min = v[i]; // Update min
  }

  cout << "Below = " << below << endl;
  cout << "Above = " << above << endl;
  cout << "Max = " << max << endl;
  cout << "Min = " << min << endl;

  return 0;   // End
}
```

program 9:

Smart Grid Load Distribution (2D Array + Functions) In a smart grid, electricity consumption is recorded across multiple regions.

Write a program to:

• Store power consumption (in MW) for 4 regions over 7 days using a 2D array.

• Create functions to:

• Calculate total consumption for each region

• Identify the day with highest total demand

• Compute overall average consumption

```
#include <iostream>
using namespace std;

// Total consumption for each region
void regionTotal(int arr[4][7])
{
   for(int i = 0; i < 4; i++)
   {
      int sum = 0;
      for(int j = 0; j < 7; j++)
      {
         sum += arr[i][j];
      }
      cout << "Region " << i+1 << " Total: " << sum
<< " MW" << endl;
   }
}

// Find day with highest demand
void highestDay(int arr[4][7])
{
   int maxDay = 0, maxSum = 0;

   for(int j = 0; j < 7; j++)
   {
      int sum = 0;
      for(int i = 0; i < 4; i++)
      {
         sum += arr[i][j];
      }

      if(sum > maxSum)
      {
         maxSum = sum;
         maxDay = j;
      }
   }

   cout << "Day with highest demand: Day " <<
maxDay+1 << endl;
}

// Overall average
void averageConsumption(int arr[4][7])
{
```

```
    int total = 0;

    for(int i = 0; i < 4; i++)
    {
        for(int j = 0; j < 7; j++)
        {
            total += arr[i][j];
        }
    }

    float avg = total / 28.0;
    cout << "Overall Average Consumption: " << avg
<< " MW" << endl;
}

int main()
{
    int grid[4][7];

    cout << "Enter power consumption (4 regions x 7
days):\n";
    for(int i = 0; i < 4; i++)
    {
        for(int j = 0; j < 7; j++)
        {
            cin >> grid[i][j];
        }
    }

    regionTotal(grid);
    highestDay(grid);
    averageConsumption(grid);

    return 0;
}
```

program 10:

Classroom Attendance System (2D Array + Functions) Teachers often track student attendance daily.

Write a program to:

• Store attendance for 5 students over 7 days using a 2D array (Use 1 for present, 0 for absent).

• Create functions to:

- **Calculate total attendance of each student**
- **Find the student with highest attendance**
- **Calculate overall class attendance percentage**

```
#include <iostream>
using namespace std;

// Total attendance of each student
void studentAttendance(int arr[5][7])
{
   for(int i = 0; i < 5; i++)
   {
      int sum = 0;
      for(int j = 0; j < 7; j++)
      {
         sum += arr[i][j];
      }
      cout << "Student " << i+1 << " Attendance: " << sum
<< endl;
   }
}

// Find student with highest attendance
void highestStudent(int arr[5][7])
{
   int max = 0, student = 0;

   for(int i = 0; i < 5; i++)
   {
      int sum = 0;
      for(int j = 0; j < 7; j++)
      {
         sum += arr[i][j];
      }

      if(sum > max)
      {
         max = sum;
         student = i;
      }
   }
```

```
    cout << "Highest Attendance: Student " << student+1 <<
endl;
}

// Overall class attendance percentage
void classPercentage(int arr[5][7])
{
    int total = 0;

    for(int i = 0; i < 5; i++)
    {
        for(int j = 0; j < 7; j++)
        {
            total += arr[i][j];
        }
    }

    float percentage = (total / 35.0) * 100;
    cout << "Overall Attendance Percentage: " <<
percentage << "%" << endl;
}

int main()
{
    int attendance[5][7];

    cout << "Enter attendance (1 = Present, 0 = Absent):\n";
    for(int i = 0; i < 5; i++)
    {
        for(int j = 0; j < 7; j++)
        {
            cin >> attendance[i][j];
        }
    }

    studentAttendance(attendance);
    highestStudent(attendance);
    classPercentage(attendance);

    return 0;
}
```

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